

ARUNKANNA S

B.E. Robotics & Automation Engineer

📍 Chennai, India | 📞 +91 80725 22040 | ✉️ mail2sarunkannaa@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [YouTube](#) | [Instagram](#)

PROFESSIONAL SUMMARY

Passionate and innovation-driven Robotics & Automation Engineering student with strong expertise in Embedded Systems, IoT, Robotics, Computer Vision, and AI-based automation systems. Experienced in developing real-world hardware and software solutions using ESP32, Raspberry Pi, Arduino, RTOS, TinyML, and Python. Skilled in designing intelligent automation systems, assistive healthcare devices, robotics prototypes, and IoT-based monitoring solutions. Strong background in rapid prototyping, CAD design using SolidWorks, and integrating AI with embedded hardware. Hackathon enthusiast with 30+ achievements across national-level competitions and project expos. Active technical community contributor, workshop conductor, freelancer, and tech content creator.

TECHNICAL SKILLS

Hardware & Core:

- Embedded Systems
- IoT System Development
- Robotics & Automation
- Hardware Prototyping
- 3D Printing

Software & Programming:

- Python Programming
- RTOS / FreeRTOS
- TinyML & Edge AI
- Computer Vision
- Face Recognition

Boards & Technologies:

- ESP32 / Arduino
 - Raspberry Pi
 - Sensor Integration
 - Web Dashboards
 - Internal Web Servers
 - Wi-Fi · BLE · AP-mode
 - Twilio API
 - CAD Design (SolidWorks)
-

EDUCATION

B.E. – Robotics & Automation Engineering *Rajalakshmi Engineering College, Chennai* | 2023 – Present

Higher Secondary School Certificate (HSSLC) *DAV Group of Schools (TNAES), Chennai* | Completed April 2023

PROJECTS

Aid for All – Smart Assistive Monitoring & Emergency Support System Developed an IoT-based assistive healthcare system for elderly people, wheelchair users, and patients requiring continuous monitoring. Integrated fall detection, GPS tracking, voice assistance, remote alerts, and caretaker communication using RTOS, internal web server, and AP-mode Wi-Fi configuration. *Technologies:* ESP32, MPU6050, DHT11, AI Thinker GPS Module, JQ6500, Twilio, RTOS, SolidWorks, 3D Printing

AI-Based Smart Assistive Wearable for Alzheimer's Patients Developed a wearable smart assistance system using Raspberry Pi and computer vision to help Alzheimer's patients recognize people, receive reminders, and improve safety. Implemented face recognition, audio assistance, caregiver alerts, geofencing, and wearable removal detection. *Technologies:* Raspberry Pi, Camera Module, Flask, Face Recognition, Audio Output, Twilio SMS

Nitroasis – AI Powered Smart Farming System Built an AI-driven smart agriculture system using IoT, TinyML, automation, and local AI assistance. Implemented real-time monitoring of environmental parameters and crop prediction using Edge Impulse TinyML on ESP32. *Technologies:* ESP32, DHT11, Soil Moisture Sensor, LDR, OLED Display, Streamlit, Ollama, Llama3, TinyML, RTOS

Twinkle – Interactive Desktop Companion Robot Designed and developed an interactive companion robot with animated facial expressions, touch interaction, audio feedback, focus timer, mini games, and Wi-Fi setup portal. *Technologies:* ESP32-C3 SuperMini, SH1106 OLED, JQ6500, Speaker, Capacitive Touch, NTP Server

Smart Helmet Safety System Developed a smart helmet system for rider safety with accident detection, drowsiness monitoring, alcohol detection, and GPS-based emergency alerts. *Technologies:* ESP32, MPU6050, IR Sensor, MAX30102, GPS Module, Alcohol Sensor, Twilio

ESP32-CAM Wireless Streaming System Developed a wireless live-streaming surveillance system using ESP32-CAM with AP mode connectivity and RTOS-based task management for real-time video streaming and image capture without internet connectivity. *Technologies:* ESP32-CAM, External Antenna, RTOS, IP5306, Battery System

Smart Home Automation for Fishes Designed a smart aquarium automation system with Bluetooth-based remote monitoring, automated feeding, and automated water maintenance. *Technologies:* Arduino Uno, HC-05 Bluetooth Module, Stepper Motor, Water Pump, Relay Module

Automated Solar Monitoring & Cleaning System Developed an ESP32-based solar panel monitoring and automated cleaning system with real-time sensor monitoring and servo-controlled cleaning mechanism. *Technologies:* ESP32, MG995 Servo, Light/Color Sensor, Web Dashboard

Robotics & Automation Projects Bluetooth Controlled Car | Wi-Fi Controlled Car | Gesture Controlled Car | Human Following Robot | Line Following Robot | Obstacle Avoiding Robot | ESP32-CAM Surveillance Car

WORK EXPERIENCE & LEADERSHIP

Campus Ambassador | GUVI

- Promoted technical learning initiatives and student engagement activities.
- Conducted outreach and technical awareness programs.

Tech Team Lead | DEVS

- Led technical teams for projects, hackathons, and workshops.
- Mentored students in embedded systems and IoT.

Technical Associate | Entrepreneurship Development Cell (EDC)

- Assisted in organizing innovation events, technical sessions, and startup activities.

Freelance Embedded Systems & IoT Developer | Self-employed

- Worked on freelance projects involving IoT systems, robotics, automation, and embedded electronics.
- Developed custom prototypes and hardware-based solutions for clients.

Workshop & Technical Event Contributor | Geekify DevFest, XploreX, P2P

- Conducted IoT workshops during Geekify DevFest.
 - Contributed to P2P Networking and XploreX technical events.
 - Assisted participants in embedded systems, IoT, and rapid prototyping concepts.
-

ACHIEVEMENTS

First Place / First Prize

- 1st Place – HARDWIRED, ABV-IIITM Gwalior (INFOTSAV'25 Hardware Hackathon)
- 1st Prize – Project Expo, S.A. Engineering College (IEEE CS Tech Savants Carnival'24)
- 1st Prize – Chitti'24 (Robo Expo), S.A. Engineering College
- 1st Place – Innovation Challenge (AID 4 ALL), Aarupadai Veedu Institute of Technology
- 1st Place – Paper Presentation, T.J.S. Engineering College (GNISTA 2K25)
- 1st Prize – CEASELESS, Rajalakshmi Engineering College (INNOVION'25)
- 1st Prize – Tech4Tomorrow Hackathon, Chennai Institute of Technology
- 1st Prize – Innovarium (30-hour Hackathon), Sri Krishna College of Engineering & Technology
- 1st Place – V-MEDITHON, VIT Chennai & Yenepoya Medical University
- 1st Place – Elevator Pitch 2024, Rajalakshmi Institute of Technology
- 1st Prize – Robo Forge, S.A. Engineering College (TRI-Tech Nexus'24)
- 1st Place – TechFusion'25 (National Project Expo), Crescent Institute of Science & Technology
- 1st Place – SPARK EXPO (Nakshatra'25), S.A. Engineering College
- 1st Place – GLYTCH Hardware Hackathon, VIT
- 1st Place – Hackathon 2K26, St. Joseph's College of Engineering
- 1st Place – ENERGATHON'26 Hackathon, SIMATS Engineering

Second Place / Runner-Up

- 2nd Place – PROTO-TRONICS S, Amrita Vishwa Vidyapeetham (EPOCH RUNNERS, Tantrotsav'24)
- 2nd Prize – Electric Wonders, Rajalakshmi Engineering College (COULOMB 2024)
- 2nd Prize – Shutter Fest, Rajalakshmi Engineering College (COULOMB 2024)
- 2nd Place – ROBOFIESTA 2K25 Hack Fest
- 2nd Position – National Entrepreneurship Challenge (IIT Bombay E-Cell)
- 2nd Place – ULTRON 9.0 Hackathon, SRM IST
- 2nd Place – Hardware Hackathon, PCB Cupid
- Runner – TechXpo'25, Jeppiaar Institute of Technology

Third Place

- 3rd Place – Mini Hackathon, Rajalakshmi Engineering College (IEEE BREADTHS)
- 3rd Place – DRESTEIN'25, Saveetha Engineering College
- 3rd Place – IQUEST'26, Rajalakshmi Engineering College
- 3rd Place – LITHOS 2K26, Chennai Institute of Technology

Special Awards & Recognition

- Tech Buddy 2.0 – ₹6000 Gift Voucher

- Best Sustainable Project – DESIGN-A-THON 24–25
 - Best Paper Award – NCARAM '24 (Aid for Communication)
-

TECHNICAL CONTENT CREATION

- **YouTube: @SAcraftworld** — robotics builds, IoT tutorials, and embedded electronics demonstrations.
- **Instagram: @arun_hobbies** — DIY engineering projects and embedded systems content.